

Lesson 11 Drive DC Motor

This lesson demonstrates on how to drive a DC motor on the expansion board to rotate at different speeds. Please note that the expansion boards B and C do not include motor interfaces.

1. Working Principle

This program enables you to set the parameters of the motor, with positive values making it move forward and negative values making it move backward. The source code of the program is located in “/home/pi/board_demo/motor_control_demo.py”.

```

31     start = False
32     print('关闭中...(closing...)')
33     board.set_motor_duty([[1, 0], [2, 0], [3, 0], [4, 0]]) # 关闭所有电机 (
close all motors)
34
35 signal.signal(signal.SIGINT, Stop)
36
37 if __name__ == '__main__':
38
39     while True:
40         board.set_motor_duty([[1, 35]]) #设置1号电机速度35(control the spe
ed of motor 1 to 35)
41         time.sleep(2)
42         board.set_motor_duty([[1, 90]]) #设置1号电机速度90(control the spe
ed of motor 1 to 90)
43         time.sleep(1)
44         board.set_motor_duty([[1, 0]])
45         time.sleep(6)
46         if not start:
47             board.set_motor_duty([[1, 0], [2, 0], [3, 0], [4, 0]]) # 关闭>
所有电机 (close all motors)
48             print('已关闭 (closed)')
49             break
50

```

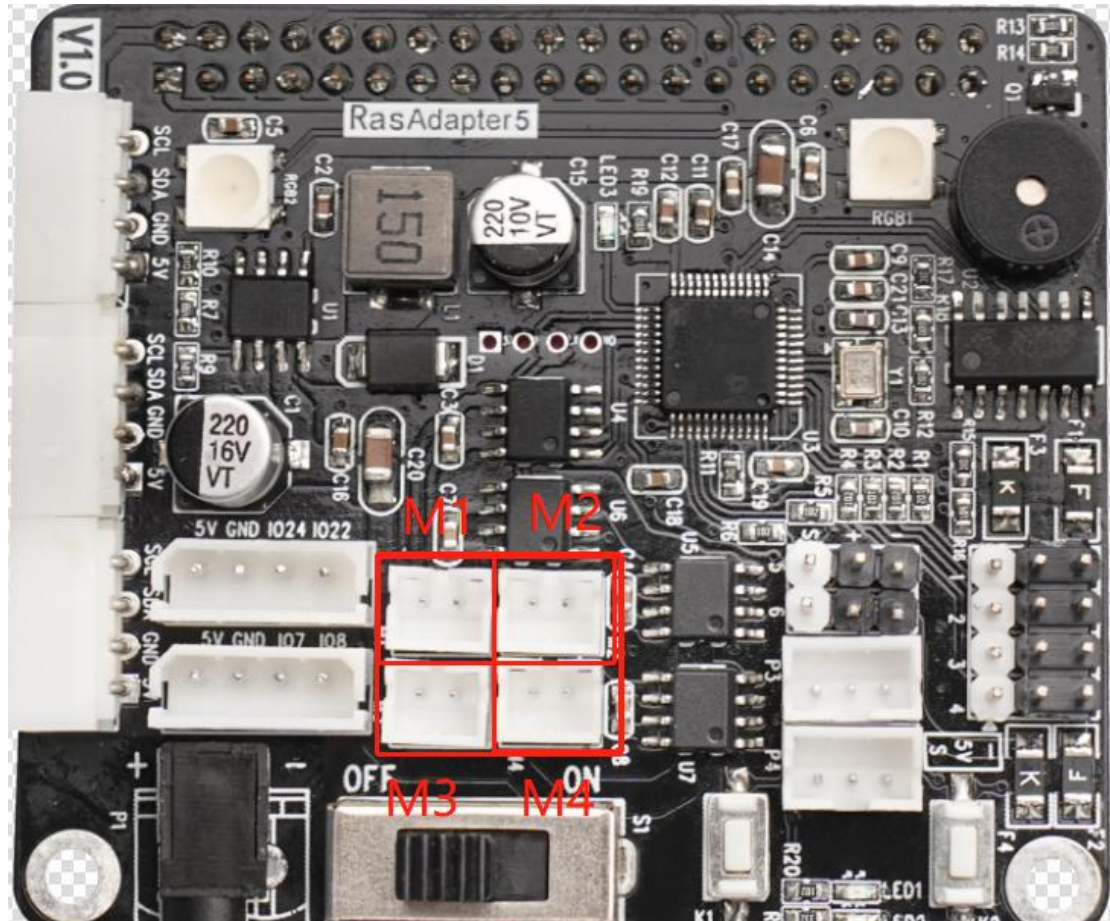
The “set_motor_duty()” function in the “Board” library is called for the control of chassis motor. Take “board.set_motor_duty([[1, 35]])” as an example:

The parameter “[[1, 35]]” sets the duty cycle of the motor. It is a nested tuple including the motor number 1 and the duty cycle 35, indicating a speed of

35.

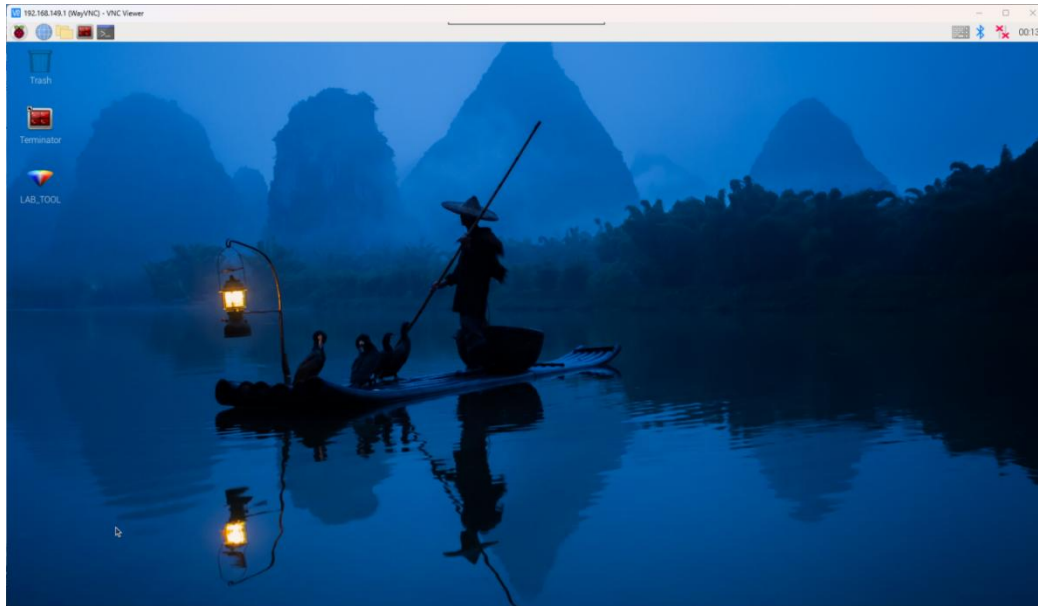
2. Getting Ready

There are 4 motor interfaces on the Raspberry Pi expansion board, as shown below. Let's test by connecting the No.1 motor interface (M1).

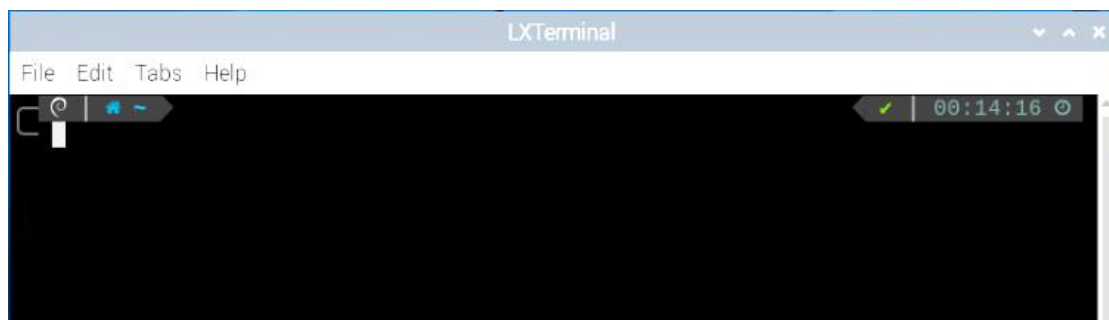


3. Operation Steps

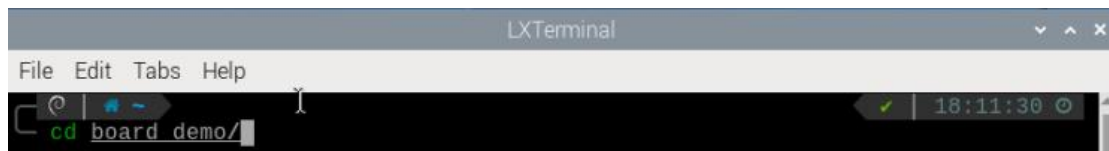
1) Turn the device on and connect the robot through the VNC remote connection tool.



2) Click  in the top left corner of the desktop, or press the shortcut key “Ctrl+Alt+T” to open the command line terminal.



3) Enter the command “cd board_demo/” to navigate to the directory where the demo program is located, then press “Enter”.



4) Enter the command of controlling the motor rotation “python3 motor_control_demo.py”, and press “Enter”.

```
pi@raspberrypi:~/board_demo $ python3 motor_contrpl_demo.py
*****
*****功能:幻尔科技树莓派扩展板, 控制直流带电机(function:Hiwonder Raspberry P
i expansion board, DC motor control)*****
*****
-----
Official website:https://www.hiwonder.com
Online mall:https://hiwonder.tmall.com
-----
Tips:
* 按下Ctrl+C可关闭此次程序运行, 若失败请多次尝试!(Press "Ctrl+C" to exit the
program. If it fails, please try again multiple times!)
-----
```

5) To close the program, press “Ctrl+c”. If the program cannot be closed successfully, repeat this operation until it exits.

4. Program Outcome

After the program is activated, the DC motor M1 will rotate clockwise at a speed of 35 for 0.2s, followed by counterclockwise rotation at a speed of 90 for 0.2s. This cycle will continue repeatedly.

5. Function Extension

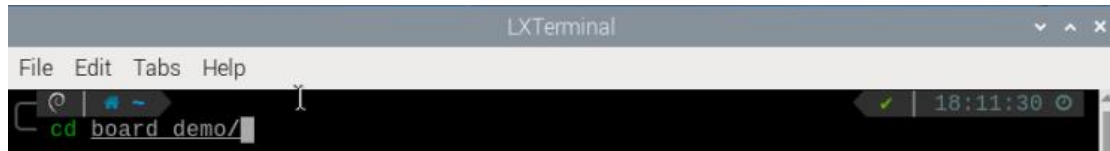
5.1 Adjust Speed and Orientation

Note: The speed range of the motor is -100 to 100. Please ensure that the values are set in this range to avoid damaging the motor.

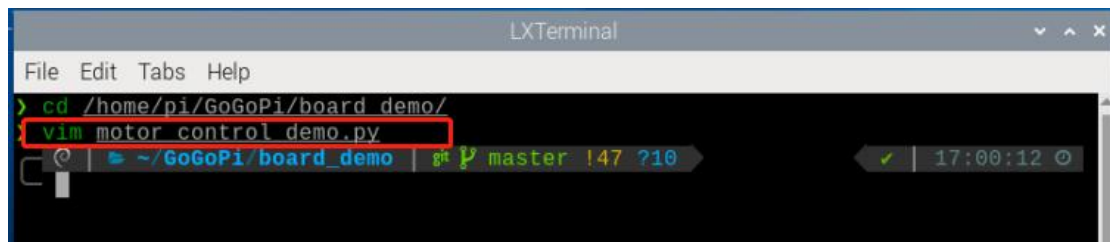
If you want to modify the speed and orientation of motor rotation, please follow the steps below.

This section demonstrates controlling the motor connected to the interface 1. It will rotate clockwise at a speed of 80 for 2 seconds, then reverse at a speed of 60 for another 2 seconds.

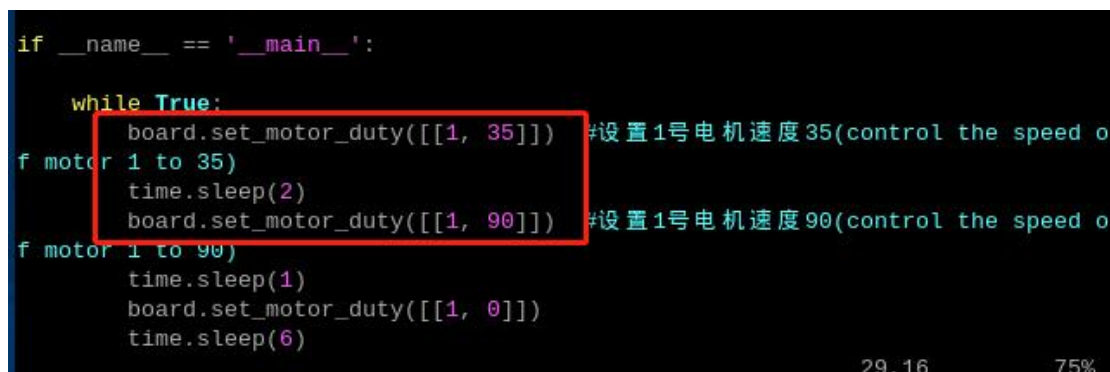
1) Enter the command “cd /home/pi/board_demo/” to navigate to the directory where the demo program is located, then press “Enter”.



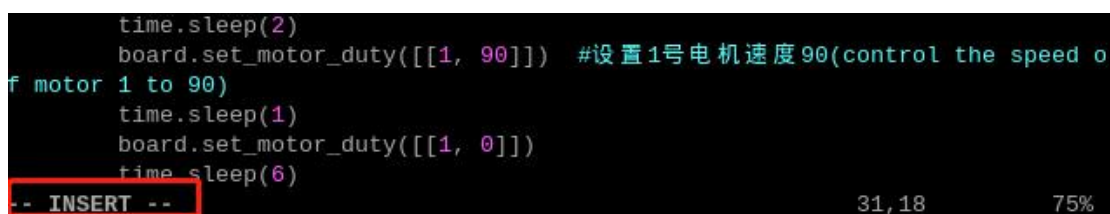
2) Enter the command of opening the program “vim motor_control_demo.py”, and press “Enter”.



3) Locate the codes shown below in the opened page.



4) Press “i” on the keyboard to enter the editing mode.



5) Modify the section highlighted in the boxed area below. Please note that when the speed value is positive, the motor rotates clockwise. The higher the value, the faster the clockwise rotation. When the speed value is negative, the motor rotates counterclockwise, and the lower the value, the faster the counterclockwise rotation.


```
signal.signal(signal.SIGINT, Stop)

if __name__ == '__main__':

    while True:
        board.set_motor_duty([[1, 80]]) #设置1号电机速度35(control the speed of motor 1 to 35)
        time.sleep(2)
        board.set_motor_duty([[1, -60]]) #设置1号电机速度90(control the speed of motor 1 to 90)
        time.sleep(2)
        board.set_motor_duty([[1, 0]])
        time.sleep(6)
    if not start:
```

6) After the modification is completed, press “Esc” on the keyboard. Then enter “:wq” (do not miss the colon before wq), and press “Enter” to save and exit the program.

```
    while True:
        board.set_motor_duty([[1, 80]]) #设置1号电机速度35(control the speed of motor 1 to 35)
        time.sleep(2)
        board.set_motor_duty([[1, -60]]) #设置1号电机速度90(control the speed of motor 1 to 90)
        time.sleep(2)
        board.set_motor_duty([[1, 0]])
        time.sleep(6)
    if not start:
:wq
```

7) After saving the program, follow the steps in “2. Operation Steps” to view its execution effects.

5.2 Adjust Motor Interface

1) The program defaults to the motor interface 1. Let’s use the modification of switching the motor interface to the motor 2 as an example. Please refer to “4.1 Adjust Speed and Orientation” to locate the code for the motor parameters, and enter the editing mode.

2) Change the motor interface from M1 to M2, as shown below:

```
if __name__ == '__main__':  
    while True:  
        board.set_motor_duty([[2, 35]]) #设置1号电机速度35(control the speed o  
f motor 1 to 35)  
        time.sleep(2)  
        board.set_motor_duty([[2, 90]]) #设置1号电机速度90(control the speed o  
f motor 1 to 90)  
        time.sleep(1)  
        board.set_motor_duty([[1, 0]])  
        time.sleep(6)
```

3) After the modification is completed, press “Esc” on the keyboard. Then enter “:wq” (do not miss the colon before wq), and press “Enter” to save and exit the program.

```
    while True:  
        board.set_motor_duty([[1, 80]]) #设置1号电机速度35(control the speed o  
f motor 1 to 35)  
        time.sleep(2)  
        board.set_motor_duty([[1, -60]]) #设置1号电机速度90(control the speed  
of motor 1 to 90)  
        time.sleep(2)  
        board.set_motor_duty([[1, 0]])  
        time.sleep(6)  
        if not start:  
:wq
```

4) After saving the program, follow the steps in “2. Operation Steps” to view its execution effects.